Rahul Bhadane

23104B0003

Problem Based Learning

Problem 6: Autonomous Drone Launch Control

An autonomous surveillance drone is allowed to launch only when mission and safety conditions are satisfied.

The drone launch signal (Y) will be generated if:

1. The mission command is received and GPS lock is available, OR
2. The operator issues an emergency launch command.

However, launch will occur only if the following constraints are satisfied:

1. The battery level must be sufficient.
2. Either the wind speed is safe OR the stabilization system is active.
3. The hardware fault signal must NOT be present.

Tasks for Students

1. Identify six Boolean variables.
    1. mission command
    2. GPS lock
    3. emergency launch command
    4. battery level
    5. wind speed
    6. stabalization system
    7. hardware fault signal
2. Define them clearly.
    1. mission command M
    2. GPS lock G
    3. emergency launch command E
    4. battery level B
    5. wind speed W
    6. stabalization system S
    7. hardware fault signal H

3. Write the Boolean expression for Launch signal (Y).

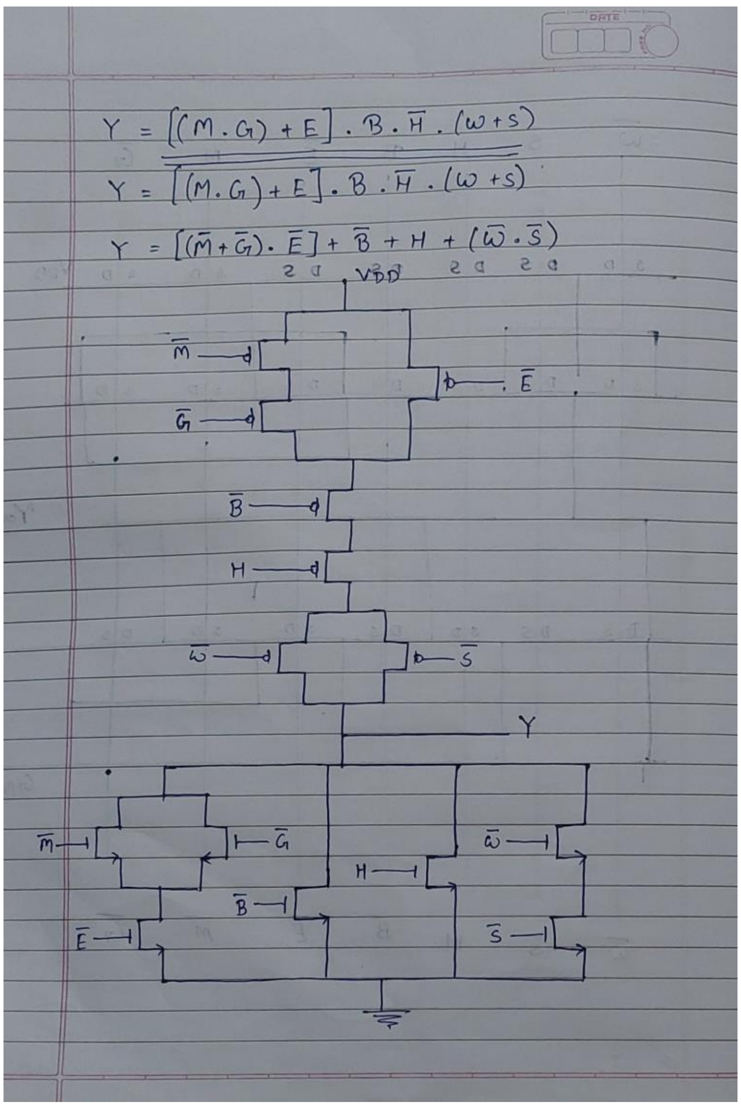
$Y = ((M \,.\, G) + E) \,.\, (B \,.\, (W + S) \,.\, H')$

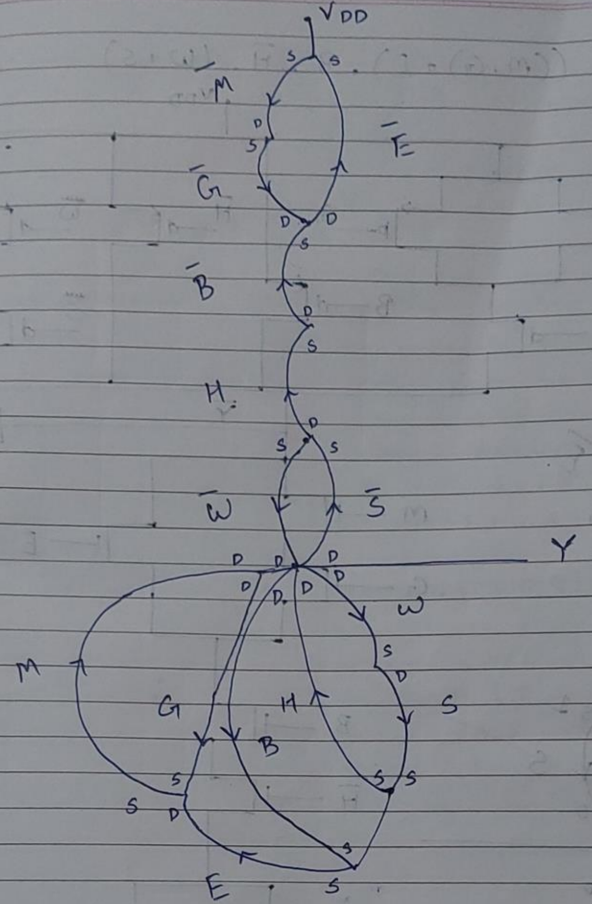
4. Simplify if possible.

$Y = ((M \,.\, G) + E) \,.\, B \,.\, H' \,.\, (W + S)$

$Y = [(M' + G') \,.\, E'] + B' + H + (W' \,.\, S')$

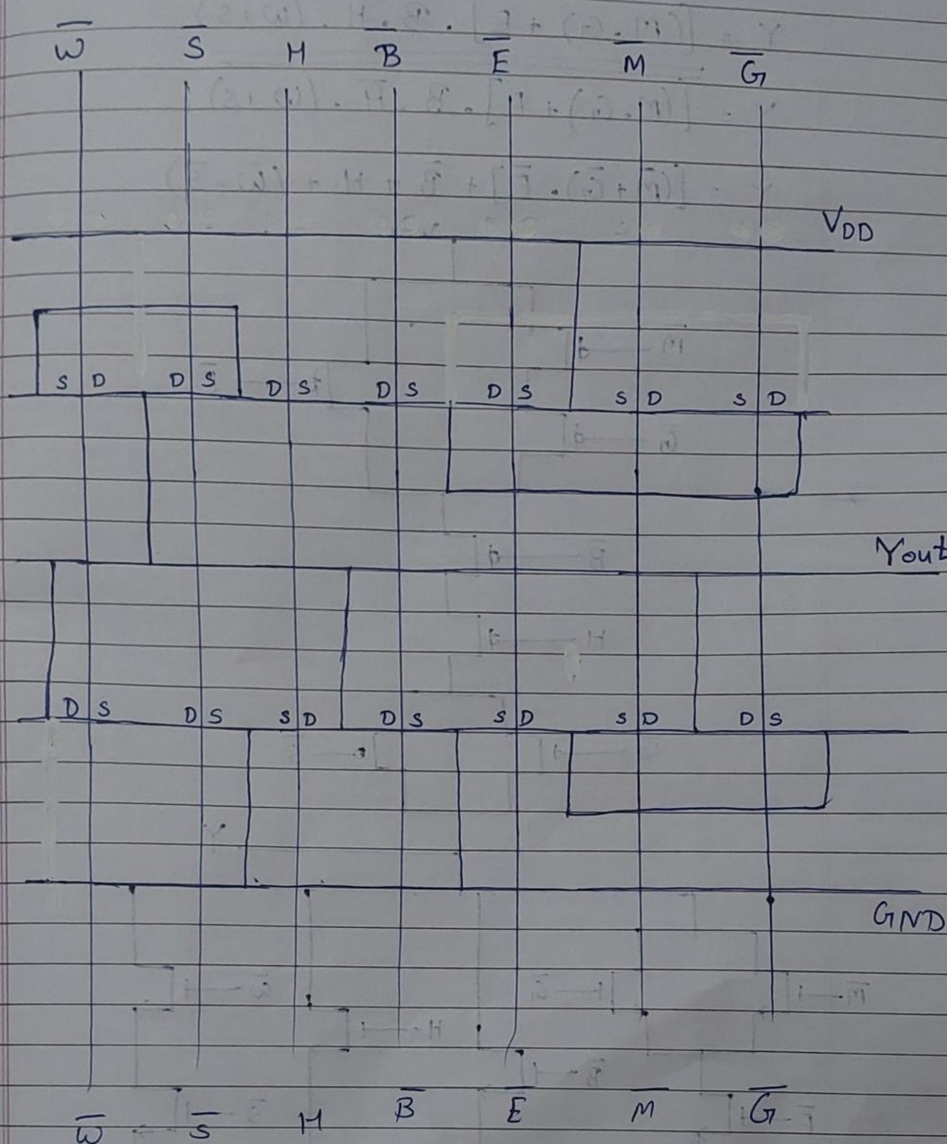
5. Implement using Static CMOS design style.





$\overline{W} \rightarrow \overline{S} \rightarrow \overline{H} \rightarrow \overline{B} \rightarrow \overline{E} \rightarrow \overline{M} \rightarrow \overline{G}$

$\overline{W}\ \overline{S}\ H\ \overline{B}\ \overline{E}\ \overline{M}\ \overline{G}$



$\overline{W}$ $\overline{S}$ H $\overline{B}$ $\overline{E}$ $\overline{M}$ $\overline{G}$
V_{DD}
S D D S D S D S D S S D S D
Yout
D S D S S D D S S D S D D S
GND
$\overline{W}$ $\overline{S}$ H $\overline{B}$ $\overline{E}$ $\overline{M}$ $\overline{G}$

6. Draw layout using Euler's path method